

ENES ARDA CANPOLAT

SOFTWARE & DATABASE ENGINEERING | PYTHON / FASTAPI / POSTGRESQL | EMBEDDED SYSTEMS

Ankara, Türkiye | Warsaw, Poland | +48 451 639 083 | +90 533 636 0070 | enesardacanpolat@gmail.com
github.com/enesardacanpolat | linkedin.com/in/enes-arda-canpolat-b49538284

PROFESSIONAL SUMMARY

Computer Science student at PJATK specializing in Software and Database Engineering (expected Feb 2028), with seven academic and personal projects across backend development, databases, embedded systems, machine learning, and software engineering. Currently developing a Python/FastAPI therapy platform backed by PostgreSQL, SQLAlchemy, Alembic, and Docker Compose. Also built interrupt-driven STM32 firmware in C, object-oriented applications in Java, Python and C#, a C++/SFML typing game, and normalized T-SQL systems with stored procedures. Uses Claude Code and ChatGPT as development tools while manually validating generated code.

TECHNICAL SKILLS

Programming Languages: C, C++, C#, Java, Python, T-SQL (SQL Server)

Backend & Databases: FastAPI, SQLAlchemy, PostgreSQL, Alembic, Docker Compose, Pydantic Settings, Jinja2, psycpgp, SQL Server, relational database design, normalization, stored procedures

Embedded Systems: STM32G4, STM32 HAL, STM32CubeIDE, UART, timers, interrupts, ring buffers, serial command parsing

Software Engineering: .NET, OOP, SOLID/DRY, dependency injection, interface-based design, exception handling, decorators, Git/GitHub

Machine Learning: Perceptrons, multi-class classification, accuracy, precision, recall, F-measure

Tools & Libraries: SFML, Claude Code, ChatGPT

Professional: Problem solving, technical documentation, intercultural communication

PROJECT EXPERIENCE

Therapy Platform (In Progress) - Backend / Database Engineering | Python | FastAPI | PostgreSQL | Docker

- Developing a FastAPI web platform with SQLAlchemy ORM models for users and therapist profiles, UUID-based entities, PostgreSQL health checks, environment-based configuration, Alembic migrations, Jinja2 templates and a Docker Compose development database.

STM32 UART & Timer Interrupt System - Embedded Systems | C | STM32 HAL

- Implemented non-blocking, interrupt-driven UART communication with ring buffers on an STM32G4; configured TIM6 for 1 Hz status updates and added a serial command parser.

Language Classification - Java | OOP | Machine Learning

- Built a multi-class classifier from scratch using one-vs-all perceptrons and 26-dimensional letter-frequency vectors; added accuracy, precision, recall, F-measure and an interactive CLI.

Personal Finance Tracker - Python | OOP | Decorators | Exception Handling

- Developed a transaction-management system with income and expense tracking, balance calculation, date and amount validation, custom exceptions and decorator-based logging.

Subscription Renewal Refactoring - C# | .NET | SOLID

- Refactored a monolithic renewal service into maintainable components using SOLID/DRY, dependency injection, interface-based design and an abstraction around an external billing gateway.

Barbershop Appointment Management System - T-SQL | Database Design

- Designed a normalized SQL Server database and stored procedures for appointments, customers, services, staff scheduling, product usage and payments.

Monkey Typer - Typing Game - C++ | SFML

- Built a configurable real-time typing game with adjustable speed, difficulty, font and font size using SFML for rendering and keyboard input.

EDUCATION

Polish-Japanese Academy of Information Technology (PJATK) - Warsaw, Poland

Oct 2024 - Feb 2028 (Expected)

BSc in Computer Science | Specialization: Software and Database Engineering

- Relevant coursework: Database Systems, Database Applications, Relational Databases, Information Systems Security, Software Engineering, Project 1 - Software and Database Engineering, Internet Technologies, Information Systems Design, Algorithms and Data Structures, Computer Networks and Java Network Programming, OOP and GUI, AI Tools, Embedded Systems, Python, C/C++, and Computer Graphics.

German Language Program - Germany

2023 - 2024

One-year German language program

- Developed independence and intercultural communication skills through international study.

TED University - Ankara, Türkiye

2021 - 2023

Industrial Design Studies - English Preparatory Program

- Completed two academic years of English preparatory study before transitioning to Computer Science.

LANGUAGES

Turkish (Native) | English (B2) | German (A2)